

Lecture 8: Clustering

September 26, 2025 (Fri)

Lecturer: Prof. Li Yu

Scribe: Chen

Recap from last lecture

Clustering

- Why clustering & what is clustering?

Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups.

- How to do clustering?

Three main elements: (1) data to be clustered, (2) similarity measurement (definition of distance), and (3) clustering algorithm (the executive procedure)

Similarity and dissimilarity measurement

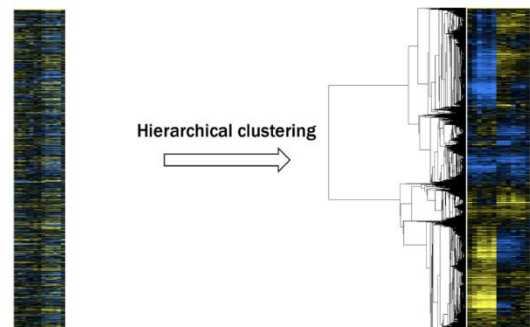
- Minkowski distance: $dist(\mathbf{p}, \mathbf{q}) = (\sum_{k=1}^m |p_k - q_k|^r)^{\frac{1}{r}}$
 - $r=1$, City block (Manhattan, taxicab, L_1 norm) distance: A common example of this is the Hamming distance, which is just the number of bits that are different between two binary vectors
 - $r=2$, Euclidean distance
 - $r \rightarrow \infty$, “supremum” ($L_{\max}$ norm, L_{∞} norm) distance: This is the maximum difference between any component of the vectors

Outline

- Hierarchical clustering
- Mahalanobis distance

Hierarchical clustering

- Originally, the data matrix (e.g. gene expression matrix) is not sorted and not clear. After hierarchical clustering, the data matrix will have a clear pattern with clusters whose data points are similar to each other. Hierarchical clustering or clustering can serve as the starting point of some further, detailed, and downstream analysis.



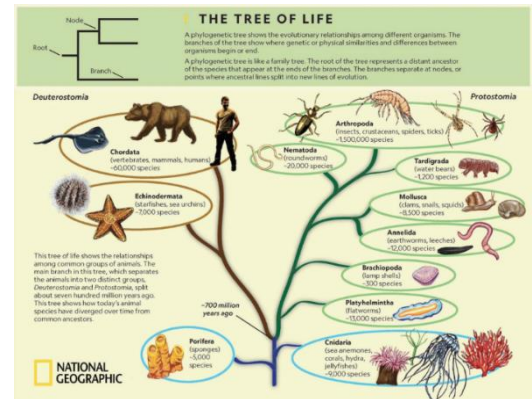
- Use similarity or dissimilarity (distance) to measure the differences

The distance can be cosine similarity, correlation, Euclidean distance, Manhattan distance, Mahalanobis distance, etc.

Intra-cluster distances are small, while inter-cluster distances are large.

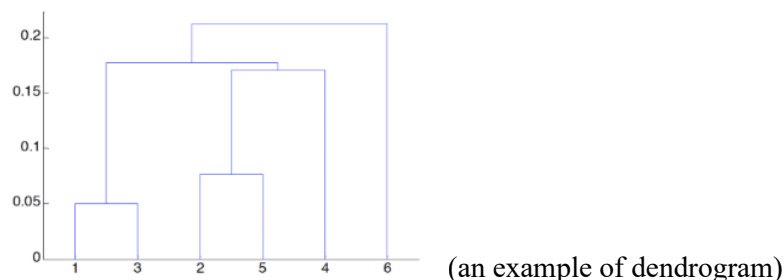
- How to identify and distinguish different groups?

We can get some hints from the tree, particularly the tree of life. At the root of the tree of life is the common ancestors; after gradually evolved, there are more branches, and the species in different branches are different from each other. Produces a set of nested clusters organized as a hierarchical tree: construct a tree, such that for each branch, the data points are similar to each other, while for different branches, the data points are different from each other.



The hierarchical tree can be visualized as a dendrogram, which is a tree-like diagram that records the sequences of merges.

The hierarchical tree may correspond to meaningful taxonomies, e.g. gene clusters, phylogeny reconstruction, animal kingdom, tree of life, etc.



(an example of dendrogram)

- Steps of hierarchical clustering
 - (1) Compute the similarity or distance matrix
 - (2) Let each data point be a cluster
 - (3) Merge the two closest clusters
 - (4) Update the similarity or distance matrix

How to update the distance matrix after merging?

The min, max, group average, or distance between centroids, etc. can be used to define the distance of two clusters.

- (5) Repeat step (3) and (4) until only a single cluster remains

- A running example:

Gene	wt	mutant_1	mutant_2	mutant_3
At4g35770	1.5	3	3	1.5
At1g30720	4	7.5	7.5	5
At4g27450	1.5	1	1	1.5
At2g34930	10	25	23	15
At2g05540	1	1	2	1

The goal is to cluster the genes.

Use correlation to define distance: $\rho_{X,Y} = \text{corr}(X,Y) = \frac{\text{cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{E[(X-\mu_X)(Y-\mu_Y)]}{\sigma_X \sigma_Y}$

- (1) Compute the distance matrix

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770	1				
At1g30720	0.9733	1			
At4g27450	-1	-0.9733	1		
At2g34930	0.9493	0.9909	-0.9493	1	
At2g05540	0.5774	0.562	-0.5774	0.4528	1

- (2) At this point, each gene is regarded as a cluster

- (3) Merge the two closest clusters:

dis(gene2, gene4) has the largest number in the distance matrix, indicating the least distance. Thus, merge gene2 and gene4, forming cluster24.

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770					
At1g30720	0.9733				
At4g27450	-1	-0.9733			
At2g34930	0.9493	0.9909	-0.9493		
At2g05540	0.5774	0.562	-0.5774	0.4528	

- (4) Update the similarity or distance matrix:

In this example, the distance is updated with the minimum distance (largest correlation) between clusters.

dis(gene1, cluster24) = max(dis(gene1, gene2), dis(gene1, gene4)) = 0.9733

dis(gene3, cluster24) = max(dis(gene3, gene2), dis(gene3, gene4)) = -0.9493

dis(cluster24, gene5) = max(dis(gene2, gene5), dis(gene4, gene5)) = 0.562

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770					
At1g30720	0.9733				
At4g27450	-1	-0.9733			
At2g34930	0.9493	->0.9733	-0.9493		
At2g05540	0.5774	0.562	-0.5774	0.4528	

- (5) Repeat the former steps of merging and updating:

gene1 and cluster24 has the least distance, thus merge gene1 and cluster24, forming cluster124.

dis(cluster124, gene3) = max(dis(cluster24, gene3), dis(gene1, gene3)) = -0.9493

dis(cluster124, gene5) = max(dis(cluster24, gene5), dis(gene1, gene5)) = 0.5774

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770					
At1g30720					
At4g27450	-1	-0.9493			
At2g34930	->0.9733	-0.9493	-0.9493		
At2g05540	0.5774	0.562	-0.5774	0.562	

cluster124 and gene5 has the least distance, merge them and form cluster1245.

dis(cluster1245, gene3) = max(dis(cluster124, gene3), dis(gene3, gene5)) = -0.5774

	At4g35770	At1g30720	At4g27450	At2g34930	At2g05540
At4g35770					
At1g30720					
At4g27450	-0.5774	-0.5774			
At2g34930			-0.5774		
At2g05540			-0.5774		

Merge cluster1245 and gene3. Only one cluster remains.

- About programming: Scikit-learn, <https://scikit-learn.org/stable/>, providing a range of functions build upon numpy, etc., which can be directly used as the plug-in functions.
- Q: In hierarchical clustering, when updating the distance matrix, do we need the original data matrix?
A: It depends on the definition of the distance between two clusters. If the definition is only related to the previous distance matrix, then the original data matrix is not required, e.g. when the distance is defined as the min or max distance of the pairs, the distance can be inferred from the distance matrix. However, if the distance definition involves the calculated coordinates based on the data matrix, the data matrix is needed, e.g. if the distance between clusters is defined as the distance between the centroids, the original data matrix is used to calculate the centroids once the cluster is updated.
- Potential project 2: Data preprocessing for the gene expression matrix
Data collecting and merging (if needed), exploration, visualization, data cleaning, get distance matrix, perform clustering

Mahalanobis distance

- When we calculate the distance, we need to consider the data distribution.
 - Example 1: The figure on the right-hand side.
When calculating the distance using Euclidean distance, then A and B have a smaller distance than A and C. However, when comparing (A and B) and (A and C) versus the entire data distribution, the distance between A and C might be smaller, since the data distribution in the direction of A to C is condensed, while that in the direction of A to B is sparse.
 - Example 2: Suppose we have two quizzes. For quiz 1, std = 10, and score of student A - score of student B = 1. For quiz 2, std = 1, and score of student C - score of student D = 1.
Q: Should the two differences (A-B) and (C-D) be considered the same?
A: No. The differences between the ranks of A and B and that of C and D could be very different.
Q: Any intuitive idea of comparing the two differences fairly?
A: Normalization. We may also use percentile or investigate there are how many standard deviations away between two points.
- Mahalanobis distance: $\text{mahalanobis}(\mathbf{p}, \mathbf{q}) = (\mathbf{p}, \mathbf{q})^T \Sigma^{-1} (\mathbf{p}, \mathbf{q})$, where Σ is the covariance matrix.

